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NHS Connect

Digital Referral & Patient Pathway Management Platform

Field	Detail
Sprint Duration	1 Week
Project Type	Digital Transformation / MVP Development
Market	United Kingdom (NHS)
Cross-Functional Team	Project Managers | Scrum Masters | PMO Analysts | Business Analysts
Delivery Approach	Hybrid
MVP Build Method	AI-assisted ("vibe coding") using Lovable
Primary Users	NHS Trust administrative & clinical staff, referrers (GPs) and patients

1. Project Overview

An NHS Trust runs outpatient services across several hospital sites and receives referrals from GPs, other clinicians and self-referral pathways into more than a dozen specialties. The Trust processes thousands of referrals every month, alongside the ongoing management of patients already on a treatment pathway.

The organisation faces a set of recurring operational problems:

- Referrals arrive through multiple, inconsistent channels — post, fax, NHS e-Referral Service, email and phone.
- Referral letters are often incomplete or poorly structured, making triage slower.
- Some clinically urgent referrals (e.g. suspected cancer) are not flagged quickly enough.
- Administrative staff manually re-key referral data across multiple systems (PAS, EMIS, SystmOne).
- Patients have little visibility of where they sit on the waiting list or what happens next.
- Managers lack a real-time view of Referral-to-Treatment (RTT) performance.
- Waiting list and capacity data exists across disconnected spreadsheets and systems.
- The Trust struggles to identify recurring bottlenecks by specialty, site or referral source.
- A high rate of Did Not Attend (DNA) appointments wastes clinical capacity and lengthens waiting lists.

The Trust wants to begin a digital transformation programme to modernise its referral and pathway management process — but does not want to build a full Electronic Patient Record (EPR)-scale system at this stage. Instead, it wants a one-week proof-of-concept MVP demonstrating how digital technology, data-informed governance and better business processes could improve the patient pathway.

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2. The Business Challenge

The current process is largely manual and linear:

Patient is referred by GP or clinician → Referral received via post/fax/email/e-Referral Service → Administrative staff log the referral manually → Referral is manually triaged for clinical priority → Referral is added to the relevant specialty waiting list → Patient waits with limited visibility → Appointment is booked manually → Reminder is sent → Patient attends or does not attend (DNA) → Outcome is recorded → Management reports are compiled later.

The Trust believes this process is inefficient. Current pain points and their business impact:

Problem	Business Impact
Paper/fax-based referral intake	Delays, lost or duplicate referrals
Manual clinical triage	Urgent and two-week-wait cases risk delay
Multiple referral channels	Inconsistent, incomplete referral data
Limited patient visibility of pathway	More calls to admin teams and PALS
Poor data quality across systems	Difficult to report accurately
Spreadsheet-based waiting list tracking	Slow management decisions; RTT breaches missed
No predictive insight into demand	Difficult to plan clinic capacity
High DNA rates	Wasted clinical capacity and longer waiting lists

3. Project Rationale

The Trust believes a digital referral and pathway platform could:

- Improve the patient experience and communication.
- Reduce avoidable calls to administrative teams and PALS.
- Improve clinical triage and referral prioritisation.
- Reduce time from referral to first appointment.
- Improve capacity planning across specialties and sites.
- Give managers real-time visibility of RTT (18-week) performance.
- Use historical data to identify bottlenecks and DNA risk.
- Create better-quality operational data to support governance and reporting.

However, before investing several hundred thousand pounds in a Trust-wide digital transformation and system integration programme, the organisation wants to test the concept.

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4. Sprint Objective

The team must design and build a working MVP that demonstrates the future-state referral and pathway journey, covering three user perspectives:

Patient side

- Submit a referral or view referral status online.
- Provide or confirm personal and clinical information.
- See the indicated clinical urgency of their referral.
- Receive a reference number.
- See the status of their pathway and an estimated wait.
- Receive appointment reminders (to help reduce DNA rates).

Staff side (administrative and clinical triage)

- View incoming referrals.
- View triage priority and clinical urgency.
- Assign or recommend the appropriate specialty/clinical team.
- Update referral and pathway status.
- Search and filter referrals and the waiting list.
- Flag and manage DNAs and cancellations.

Management

- View referrals and RTT performance dashboard.
- Monitor KPIs, including 18-week RTT compliance.
- Identify bottleneck specialties and sites.
- Analyse referral trends and demand.
- Identify recurring causes of backlog and breach.

5. The MVP

The team is not expected to build a production-ready NHS system, and the MVP must not use real patient data. The MVP should demonstrate the concept and value proposition through six core features.

Feature 1 — Digital Referral Intake

A referrer/patient-facing form capturing: specialty (e.g. Cardiology, Orthopaedics, ENT, Dermatology, General Surgery, Gynaecology, Other), patient reference number, referral reason/description, clinical priority indicated by the referrer, preferred site/location, and contact preference.

Feature 2 — Automated Triage Categorisation

The system should use predefined rules or AI to classify the referral. For example, a referral describing red-flag symptoms should be classified as Priority: Two-Week Wait, Recommended Team: Suspected Cancer Pathway. A rules-based approach or AI-assisted classification is acceptable for the MVP — a sophisticated clinical model is not required, and the team must not attempt genuine clinical diagnosis.

Feature 3 — Pathway Prioritisation & RTT Tracking

The system should calculate a suggested priority and track progress against the 18-week Referral-to-Treatment standard. Defining the business rules behind prioritisation is an important Business Analyst activity.

Priority	Example
Two-Week Wait	Suspected cancer symptoms
Urgent	Rapidly worsening condition
Routine	Stable, non-urgent condition
Planned / Surveillance	Routine follow-up review

Feature 4 — Waiting List & Performance Dashboard

The management dashboard should show executive KPIs: total referrals, open referrals, referrals breaching the 18-week RTT standard, average wait time, two-week-wait compliance, DNA rate, referrals by specialty, site and referral source, and patient satisfaction.

6. Dataset

A synthetic operational dataset is provided separately as an accompanying spreadsheet workbook (not reproduced in this brief), covering referral records and patient pathway records for the Trust. It is designed so teams can begin exploratory analysis on Day 1 without needing to source or handle real patient data — no genuine patient-identifiable data should be used at any point in this exercise.

Teams may also reference genuine UK contextual datasets where useful, including NHS England RTT waiting times statistics, the NHS Digital / NHS England Open Data portal, GOV.UK Open Data, the Office for National Statistics (ONS) and NHS Model Health System benchmarking data. Teams should not spend sprint time hunting for perfect external datasets — the primary operational dataset is supplied in advance.

7. Team Structure

Each project team should ideally contain the following cross-functional roles, working together as one integrated PMO:

Project Manager

- Project Coordination
- Project Charter development
- Stakeholder Analysis - Stakeholder register
- Manage Documentation Repository
- Delivery coordination and final presentation.

Scrum Master

- Project Planning and Sprint Management
- JIRA Board Creation
- Facilitate Daily standup

PMO Analysts

- RAID log / risk register maintenance and governance reporting.
- KPI definition, dashboard build and trend analysis.
- Consolidating insights and recommendations for the Final Project Report.

Business Analysts

- Current-state (as-is) and future-state (to-be) process mapping.
- PRD, user stories development
- Acceptance Criteria.

Everyone: AI / Vibe Coding Lead

- Leads the AI development tool (Lovable).
- Front-end MVP build and connecting the sprint dataset.
- Prompt engineering and UI/UX implementation.

Everyone is encouraged to participate in the build rather than creating a separate “developer” silo — this better simulates a modern cross-functional NHS digital transformation team.

- Product Backlog
- User Stories
- Acceptance Criteria

Final

- 10–15 minute presentation
- 5-minute live MVP demonstration

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9. The Employability Dimension

The real value of this exercise is that participants finish the week with a concrete interview story. Rather than saying "I completed a Project Management course," a candidate can say they worked in a cross-functional Agile PMO to deliver a digital transformation MVP for an NHS Trust — working with Business Analysts to map referral pathways, defining KPIs and RTT reporting, maintaining governance and risk logs, and using AI-assisted development to build a working referral management platform within a five-day sprint.

Interview Questions This Project Prepares Teams For

Project Management

- Tell me about a project you managed.
- How did you manage competing priorities?
- How did you manage scope?
- What risks did you identify?
- How did you handle a blocker?
- How did you measure project success?

PMO & Governance

- How do you maintain a RAID log and keep it useful?
- How do you track and report progress to stakeholders?
- How did your reporting influence a business decision?

Business Analysis

- How do you gather requirements?
- How do you distinguish a requirement from a solution?
- Show me how you would write a user story.
- How would you map an as-is and to-be process?
- How do you manage conflicting stakeholder requirements?

AI / Digital Transformation

- How have you used AI in your work?
- How would you use AI to accelerate product development?
- How do you validate AI-generated outputs?
- What are the risks of using AI in a healthcare business environment?